

# SAMPLING-UNIT AND COMPOSITION SENSITIVITY OF AN AUDIO DEEPPFAKE LEADERBOARD

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## ABSTRACT

Leaderboard comparisons often treat correlated evaluation trials as independent. On ASVspoof 2021 DF, a matched trial-i.i.d. procedure excludes zero for five of six organizer-baseline pairs, whereas speaker-attack product perturbation and a coherent marginal-sum sensitivity covariance exclude zero for none. A pre-access-frozen single-source SpoofCeleb test, where between-source composition cannot move, changes the corresponding count from 6/6 to 3/6; a disclosed post-result decomposition gives 5/6 under speaker-only and 3/6 under attack-only perturbation. Across 12 registered score-blind composition policies, 6/12 within-generation orderings reverse but 0/16 cross-generation orderings do; a disclosed post-failure search finds reversal witnesses as small as  $r_{TV} = .010218$ . Under an imposed low-EER random-effects DGP, nominal-95% coverage is .879/.880/.883, refuting universal calibration under that DGP. Thus the robust fixed-score conclusion is coarse generation separation, not within-generation ranking. These are evaluation-design sensitivity results, not population confidence claims.

**Index Terms**— Anti-spoofing, audio deepfake detection, evaluation methodology, clustered bootstrap, ranking uncertainty

## 1. INTRODUCTION

In 2021 the ASVspoof organizers tested every pair of submitted systems for significance, applied a Holm–Bonferroni correction, and published the deepfake task’s systems in four groups, noting that the top two were not separated [1]. We found no uncertainty analysis in three subsequent benchmark reports: the journal overview [2], ASVspoof 5 [3], and the recent multi-benchmark leaderboard [4] contain none of *significance test*, *confidence interval*, *bootstrap*, *p-value* or *standard error*. The sub-point margins on that modern leaderboard are therefore reported without an uncertainty test; a 2026 benchmark of self-supervised models likewise ranks point EERs without an interval, error bar or test [5]. The 2026 edition adds a calibration study instead; calibration

establishes whether a score means what it says, not whether two systems differ.

That matters more than it would if the analyses the field did run had represented the dependence. ASVspoof 2017 reported 95% EER confidence intervals and read their overlap [6], and 2021 tested every pair [1]; both use [7], and the practice then lapsed. Both variants of the cited test [7] treat trials within a class as independent Bernoulli draws; the dependent form accounts for two systems sharing trials, not for trials sharing speakers and attacks. ASVspoof 2021 DF contains 533,928 evaluation trials but only 93 speakers and 110 attacks, and per-attack EERs for SSL-AASIST span 0.4–10.7%. Speaker ICC on bona-fide scores is 0.15–0.77, so repeated trials plainly do not supply independent information. ICC alone does not establish paired error correlation or an EER design effect; those are measured directly below by matched perturbations of the joint scores and then checked externally at family level on ASVspoof 5.

We audit two evaluation-design choices on fixed released scores: the perturbation unit and the observed source/task weighting. The primary question is whether pairwise zero-exclusion decisions survive those choices; we propose no new bootstrap and claim no identified population sampling law. On 21DF the organizer-family count changes from 5/6 to 0/6, and on pre-access-frozen single-source SpoofCeleb it changes from 6/6 to 3/6. A disclosed post-result factor decomposition localizes the latter pattern to 5/6 under speaker-only and 3/6 under attack-only perturbation. Composition policies then localize the stable conclusion: cross-generation separation survives, whereas within-generation ordering does not. Coverage stress and external families bound rather than widen that fixed-data claim.

**Scope.** The primary pool comprises eight author or organizer releases with high provenance, not the universe of public re-scores. We analyze all eleven Speech DF Arena re-scores [4] separately. No full-21DF “resolved” label transfers to new speakers or attacks: four source/task blocks have only 4–6 speakers each. Bands and composition radii are descriptive procedure outputs on fixed scores, not population confidence or safety claims.

## 2. RELATED WORK

**This benchmark’s own reporting.** ASVspoof 2021 pairs systems with the test of Bengio and Mariéthoz [7] under Holm correction and publishes group structure [1]; Wang and Yamagishi [8] showed that changing only the training seed moves ADD rankings—a variance component our intervals do *not* include.

**Clustered resampling.** Poh, Martin and Bengio [9] showed that biometric uncertainty is governed by users rather than sample count; Wu et al. carried user-aware resampling into speech [10] and developed a paired correlation-corrected two-system test [11]. Bolle et al. proposed a subsets bootstrap [12], while Fogliato et al. found undercoverage for dyadic 1:1 matching [13]. Our trials instead have one speaker and, for spoof, one attack: two crossed factors [14], a delicate design [15, 16] distinct from nested metrics [17]. Speech bootstrap intervals also include utterance- and block-based forms [18, 19]. Our contribution is the audited fixed-data result: after EER threshold refitting and all-pair multiplicity are aligned, outputs change with the declared perturbation unit. We claim neither cell-level agreement with the published greyscale figure nor population confidence.

**Benchmarks that cannot resolve what is claimed on them.** Zhuang et al. [20] compute a paired-binomial detectable effect for quantization benchmarks, treating items as independent—the assumption under test here. In audio deepfakes, aggregation weights change conclusions [21], channel presentation can dominate model choice [22], and other audits question representations [23] and corpus overlap [24]. None gives pairwise composition-reversal witnesses or names a dependence unit for uncertainty. Simultaneous rank intervals are established [25]; adjacency is score-selected, so we control multiplicity over the whole family.

## 3. METHOD

### 3.1. Identified object: fixed-data procedure sensitivity

Two deterministic systems on a frozen trial list have *no* sampling uncertainty; an interval exists only after declaring what was sampled. The organizers’ test treats trials within class as independent draws [7]. Our original working target instead treated speakers and attacks as exchangeable with a represented evaluation design. Metadata falsify a global version of that design: source/task stratification leaves almost no empty cells, yet the VCC blocks contain only 4/4/4/6 spoof speakers. We therefore do not replace the organizers’ population claim with another unidentified one. The identified object is how paired  $\Delta$ EER outputs on the fixed score set change under declared trial-i.i.d. and speaker–attack perturbation laws; confidence language is reserved for a future target with a defensible sampling law and replication.

### 3.2. Clustered bootstrap and simulation-checked variants

Speakers and attacks are resampled independently with replacement and trial weights multiply the two counts; bona fide trials carry no attack label and so receive speaker weights only. The primary output is the two-way percentile bootstrap. Exact delete-one refits separately compute  $\Sigma_s + \Sigma_a - \Sigma_{sa}$  over 93 speakers, 110 spoof attacks and 1,062 observed spoof speaker $\times$ attack cells [14]. Its former pairwise floor  $\max(V_s + V_a - V_{sa}, V_s, V_a)$  did not define a joint covariance and is retired from simultaneous labels. The replacement corroborator uses the single PSD matrix  $\Sigma_+ = \Sigma_s + \Sigma_a$ : it dominates both marginals and the raw exact-cell matrix in Loewner order, and the same  $\Sigma_+$  supplies every pair SE and the Gaussian max- $t$  critical value ( $q = 2.878$ ). It deliberately retains the cell overlap and is a coherent marginal-sum sensitivity covariance, not an exact multiway estimator or coverage claim. Product and coherent outputs agree on all 28 zero-exclusion labels (18/28; organizer 0/6). A smooth influence-function CRVE failed its prospectively frozen internal finite-refit gates on every pair and is excluded.

### 3.3. All-pair multiplicity for procedure bands

From  $B=5000$  clustered replicates we form sup- $t$  simultaneous procedure bands [26]: the critical value is the 95th percentile of  $\max_p |\Delta_p^{(b)} - \bar{\Delta}_p|/s_p$  over replicates, applied to every pair. Rank sets are derived mechanically from those bands. Because the required sampling law is not identified for full 21DF, neither is called a population confidence set. A bootstrap rank percentile is separately released as a stability interval.

**Reconstructing the cited test.** The overview does not disclose which Bengio–Mariéthoz variant produced its greyscale matrix, so we make no cell-level attribution. With each system’s non-interpolated EER threshold selected on the evaluation set, the independent and paired-decision reconstructions separate the same five of six baseline pairs. Holm is step-down, but the result is family-size independent: in the independent reconstruction the weakest significant pair has  $p = 9.7 \times 10^{-6} < 0.05/561$ , while the sixth has  $p = 0.217 > 0.05$ . Because the cited derivation assumes thresholds fixed outside the test set, these are reconstructions of its adaptation to EER rather than exact finite-sample tests.

### 3.4. Coverage check

A bivariate random-effects model generates scores on the real trial index, preserving sparse incidence while imposing the exchangeability law that source/task metadata later reject. A frozen 48-cell grid passes all 24 organizer-like cells (coverage ranges .928–.957 exact-cell and .946–.968 product) but fails at low EER: the three candidate procedures have minima .845/.845/.855 and all fall below .90 in

11/24 cells. An oracle-SD diagnostic identifies asymmetric non-pivotality rather than average variance scale. The pre-registered universal-calibration reading is therefore Refuted. This diagnoses procedure behavior under the imposed DGP, not adequacy of that DGP for 21DF.

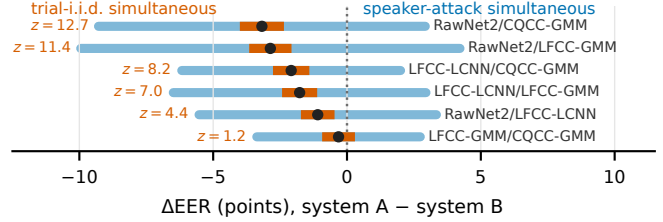
### 3.5. Composition sensitivity on fixed scores

Three bona-fide policies (empirical, source-balanced, source-speaker-balanced) and four spoof policies (empirical through attack-family-cell balanced) form a frozen 12-policy factorial. All trial weights remain positive and class-normalized. For each pair we report the policy range and registered empirical-to-policy paths;  $r_{TV}$  is the larger class-conditional total-variation shift. These are deterministic estimands, not confidence intervals. After two failed numerical implementations, a separately frozen post-failure search may only exhibit constructive upper bounds: a witness must reverse under count-canonical, direct-trial and tie-safe EER evaluators. Search failure is not stability; a reported bound is not a global minimum.

## 4. EXPERIMENTS

**Setup.** The primary pool contains eight high-provenance ASVspoof 2021 DF releases: four self-supervised systems (XLSR-Mamba [27], XLS-R+SLS [28], XLSR-Conformer [29], SSL-AASIST [30]) from their authors and four organizer baselines (RawNet2, LFCC-LCNN, LFCC-GMM, CQCC-GMM) from the challenge key package. We use the 533,928 eval-phase trials of 611,829 released. Pooled EERs run 1.88–2.85 for the self-supervised systems and 22.38–25.56 for the baselines, reproducing published values where they exist. Pairwise intervals use  $B=5000$  clustered replicates (seed 20260817); the 48-cell coverage stress test uses  $B=500$  per replicate over  $R=1000$  replicates per cell (seed 20260824).

**Primary 21DF sensitivity result** (Fig. 1). Both published-test reconstructions separate the same five of six organizer pairs; we claim no cell-level agreement with the greyscale figure. A matched all-28-pair max- $t$  bootstrap refits the same non-interpolated weighted EER in every replicate: trial-i.i.d. perturbation gives 26/28 and 5/6 organizer exclusions, against 18/28 and 0/6 under speaker-attack product weights. Coherent marginal-sum covariance gives the same 18/28 and 0/6. The matched threshold analysis removes threshold treatment as an explanation. A disclosed post-failure control also gives 18/28 and 0/6 while restoring each class-by-source/task mass exactly; it shows that movement in that mass is not necessary for the contrast, but is a constructive robustness control rather than a prospective causal decomposition. The widest 3.18-point gap has product band  $[-9.27, 2.91]$  and coherent band  $[-8.76, 2.40]$ .



**Fig. 1.** The six baseline pairs, with matched all-28-pair max- $t$  bands. Narrow bars: simultaneous trial-i.i.d. output; wide bars: simultaneous speaker-attack output. The independent Bengio-Mariethoz reconstruction’s  $z$  is shown at left and gives the same five verdicts as the narrow bars. Dots: point estimates. All six speaker-attack bands include zero.

**Pre-access single-source check.** On the complete 91,130-trial SpoofCeleb grid [31], four fixed off-domain detectors give 6/6 trial-i.i.d. versus 3/6 product exclusions under a rule frozen before access. AASIST, SLS, SSL-AASIST and XLSR-Mamba have point EERs 57.93%, 24.51%, 26.72% and 27.58%, respectively; this check concerns a weakly transferred off-domain family, not high-performing in-domain SpoofCeleb systems. Every base item has one bona-fide and nine spoof renditions under a single source label. Source composition is therefore fixed by design, although product draws still vary speaker and spoof-attack multiplicities; between-source movement is not necessary for the contrast. A disclosed post-result decomposition gives 5/6 speaker-only and 3/6 attack-only exclusions, the latter reproducing the joint verdict vector: under these procedures attack-level perturbation alone is sufficient for all three changes, while speaker-only is sufficient for one. Gate 2 later found that the original seal omitted the actual scorer endpoint. A disclosed post-result rerun from clean, hash-bound commits reproduced three score files byte-for-byte; XLSR-Mamba differed by at most  $1.44 \times 10^{-6}$ , with all EERs, decisions and the registered endpoint unchanged.

**What the benchmark supports** (Table 1). All six organizer pairs straddle zero under product and coherent marginal-sum perturbations despite a 3.18-point span. RawNet2-CQCC-GMM, LFCC-LCNN-LFCC-GMM and LFCC-LCNN-CQCC-GMM become zero-excluding in leave-one-corpus-out refits, so the organizer result is not composition-robust. The 16 cross-generation gaps are 19.5–23.7 points and survive every deletion. Two modern zero-exclusions at 0.97 and 0.94 points are withdrawn because low-EER stress is anti-conservative and both disappear after source deletion; adjacent modern margins are only 0.03, 0.36 and 0.58 points.

**Composition robustness.** Under the 12 registered score-blind policies, 6/12 within-generation orderings reverse and 0/16 cross-generation orderings do; only one registered path reverses at  $r_{TV} \leq .10$ , yielding the frozen classification *Local*

**Table 1.** Within-generation pairs: empirical gap and product-perturbation pointwise percentile half-width (EER points; the simultaneous bands are reported in the text), plus the smallest verified constructive  $r_{TV}$  upper bound. A bound is a found reversal, not a minimum or confidence limit.

| pair       |           | gap   | half-w. | $r_{TV} \leq$ |
|------------|-----------|-------|---------|---------------|
| XLSR-Mamba | XLS-R+SLS | 0.032 | 0.452   | 0.011         |
| XLSR-Mamba | XLSR-Conf | 0.389 | 0.619   | 0.344         |
| XLSR-Mamba | SSL-AAS   | 0.968 | 0.555   | 0.246         |
| XLS-R+SLS  | XLSR-Conf | 0.357 | 0.752   | 0.172         |
| XLS-R+SLS  | SSL-AAS   | 0.935 | 0.561   | 0.434         |
| XLSR-Conf  | SSL-AAS   | 0.578 | 0.674   | 0.166         |
| RawNet2    | LFCC-LCNN | 1.094 | 2.868   | 0.069         |
| RawNet2    | LFCC-GMM  | 2.864 | 4.638   | 0.094         |
| RawNet2    | CQCC-GMM  | 3.180 | 3.952   | 0.112         |
| LFCC-LCNN  | LFCC-GMM  | 1.770 | 3.074   | 0.087         |
| LFCC-LCNN  | CQCC-GMM  | 2.086 | 2.636   | 0.104         |
| LFCC-GMM   | CQCC-GMM  | 0.316 | 1.996   | 0.034         |

*composition fragility.* The later constructive search cannot revise that classification: it verifies 729 directions and bounds all 12 within-generation reversals from above, five at  $r_{TV} \leq .10$ . XLSR-Mamba versus XLS-R+SLS reverses at .010218. All three EER evaluators agree, but the delta at that witness is tiny; the result is local sensitivity, not practical superiority of either system. No cross-generation witness was found, which is a search outcome rather than proof of global stability.

**Independent re-score layer.** Treating the eleven complete Speech DF Arena re-scores as a separate 55-pair provenance stratum changes the zero-exclusion count from 52 under trial-i.i.d. perturbation to 38 under speaker-attack perturbation; the median pointwise-width ratio is 8.45. These are descriptive sensitivity outputs, not formal Arena confidence bands.

**Speakers are not exchangeable either.** The three bona-fide sources contain 67/12/14 speakers; corpus-grouped delete-one influence contributes 9–48% of its finite- $A$  sum of squares (median 26%). Against a 2.2% permutation null, all 12 pairs survive Holm ( $p_{\text{Holm}} \leq .041$ ): treating corpus as exchangeable leaves speaker bands anti-conservative. Leave-one-corpus-out exact-cell refits preserve all 16 cross-generation verdicts, remove both within-SSL resolutions and add three baseline ones. Thus block separation survives provenance changes, while every within-generation resolution is conditional on them.

**External family-level check.** On ASVspoof 5 [3], four frozen detectors over all 680,774 trials meet both registered criteria: at least one trial-i.i.d. exclusion disappears under role-stratified speaker-attack perturbation, and median width inflation is  $27.27\times$ . This is an external fixed-family sensitivity check, not pair-level replication or population inference.

**Measured width inflation.** On the real scores, simultaneous product bands are  $4.13\text{--}11.36\times$  wider than matched trial-i.i.d. bands over all 28 pairs (median  $8.14\times$ ). Under the

fitted organizer-like Gaussian DGP on the real incidence, a witnessed  $R=200$  run gives 18.5% trial-i.i.d. coverage (Wilson 13.7–24.5%) versus 97.5–98.0% for the clustered candidates (combined Wilson endpoints 94.3–99.2%). This calibrates procedures under the imposed DGP, not that DGP as a model of 21DF.

## 5. DISCUSSION

**For reporting, and limits.** Plans, deviations, hashes and receipts are at [github.com/rvirgilli/asvspoof2021-df-clustered-uncertainty](https://github.com/rvirgilli/asvspoof2021-df-clustered-uncertainty). Legacy SSL-AASIST/AASIST NPZs lack historical run provenance, and the historical freeze lacks an independently verifiable timestamp. Report speaker and attack incidence plus composition, not trial counts: bands require a sampling law and coverage; radii require a policy family and witness. We withdraw full-21DF population labels, modern rank-confidence claims and formal Arena bands. Zero-straddling is sensitivity, not equality.

## 6. CONCLUSION

On fixed released scores, matched 21DF perturbations change organizer-family exclusions from 5/6 to 0/6, and a pre-access-frozen single-source SpoofCeleb check changes them from 6/6 to 3/6. Because SpoofCeleb has one source, between-source movement cannot explain the latter contrast; a post-result decomposition shows that attack-only perturbation reproduces its 3/6 verdict vector. Registered composition policies reverse 6/12 within-generation orderings but 0/16 cross-generation orderings. Across the registered policies, corpus deletions and searched directions, the robust conclusion is coarse generation separation, not within-generation ranking. Population confidence requires independently sampled units and a defensible acquisition law.

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